

Weekly Discussions

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How these work

Six of the seven weeks carry a discussion. It is stage 2 of the **peer-engagement loop**, and it is where the week's concept gets stated in your own words and then tested against how your teammates read it. **Week 5 has no discussion**: it is the project midpoint, so the loop runs tool-only and the writing energy goes to the project.

You stay in the **same project team** for the whole term. The people who respond to your Week 2 post are the people you build the **course project** with, so the discussion is also how your team learns to give each other useful technical feedback before the stakes get high.

What completion requires

Four stages. All four are required for credit.

1. Work through the module's **interactive tool** and pass its built-in mastery check. Module 3 has no tool, so the written post stands in for it.
2. Post your own answer to the week's prompt in your team's D2L discussion.
3. Respond to **every member of your project team**, one response each.
4. Reply to the responses left on your own post, either revising your claim or defending it.

Missing any stage is a **Not-Yet** for that week. Because stages 3 and 4 depend on your teammates having posted first, the stages are time-bound within the week and cannot be made up after the loop closes.

Timing. Initial post by **midweek**; peer responses and replies by **the end of the week**. The exact days for each offering are posted on D2L.

The quality bar

Length is a floor for substance, not the target. A short post that makes a real claim beats a long one that does not.

Component	Length	What it must contain
Your post	150 to 250 words	Claim: answers the prompt directly with a position, not a summary. Evidence: at least one concrete result from your own work that week (a number, a figure, an output from the tool, the homework unit, or a project milestone). Reasoning: states how that evidence supports the claim. Limits: names one condition under which your claim would fail, or one thing you could not determine.
Response to each teammate	75 to 125 words each	Quotes or paraphrases a specific claim in your teammate's post, then does at least one of: extends it with evidence or a case they did not consider; challenges it with a stated reason and an alternative; connects it to another module, to a differing result of your own, or to a project milestone.
Reply on your own post	50 to 100 words per thread	Engages what was actually raised. Either revise your claim and say what changed it, or defend it with reasoning that addresses the objection.

What does not pass

- Restating the prompt, or summarizing the reading without taking a position.
- A post with no evidence from your own work that week.
- A post that names no limit.
- Agreement without addition. "I agree, good point" is not a response.
- A response that restates your teammate's claim back to them.
- Responding to fewer than all of your teammates.

The discussions

Discussion 1 · Week 1 · M1 Foundations

Introductions and what you want to be able to do

This week only, the evidence requirement is waived: the course has not produced any of your own results yet. Every other requirement applies.

Prompt. Introduce yourself to your project team: your background, what brought you to this course, and what you currently use Python for. Then commit to something specific. Of the four problem types this course covers (regression, classification, clustering, and dimensionality reduction), name the one you expect to find hardest and say why. Close with one thing you want to be able to do by Week 7 that you cannot do now.

A strong post. Give a reason rooted in your own experience rather than a general statement that a topic is difficult, and state a Week 7 goal specific enough that you could tell in Week 7 whether you hit it.

Discussion 2 · Week 2 · M2 Bias-Variance and Regularization

The tradeoff in your own words

Prompt. Describe the bias-variance tradeoff without using the words “bias” or “variance.” Ground it in your own runs of the [Week 2 explorer](#): report the polynomial order and the ridge penalty you settled on, and say what told you to stop there. Then the harder question. On this NHANES cohort the gain from higher polynomial order is small, and the dramatic underfit and overfit contrast you see in textbooks is partly an artifact of synthetic data. Does that make the tradeoff less important to understand, or more? Take a position.

A strong post. The two-word constraint is there to force you to explain the mechanism rather than name it. Expect your team to split on the second question. That disagreement is the point, so respond to it rather than around it.

Discussion 3 · Week 3 · M3 Probability, Likelihood and MLE

What likelihood buys us

Module 3 has no interactive tool, so this post carries the module’s peer-engagement loop on its own. It is held to a higher standard than the others.

Prompt. Under Gaussian noise, maximum likelihood recovers exactly the least-squares solution you already had in Week 1. So what did you gain? Name something specific that the likelihood formulation gives you which least squares alone cannot, and show it with a result from your own Unit 3 or milestone 1.2 work. Then state one claim you can now make about your Part 1 fit that you could not have made after 1.1.

A strong post. Resist the answer “it is more principled.” Point at a quantity that did not exist before you wrote down a likelihood, and show its value from your own run.

Discussion 4 · Week 4 · M4 Bayesian Inference

How the posterior contracts

Prompt. Using the [Week 4 Bayesian updating explorer](#), describe what happened to your posterior as you fed it more data. Report the specific point at which your prior stopped visibly mattering. Then take a position: does the fact that the prior washes out make the choice of prior less important, or more important? Defend it. If you think the answer depends on something, name what it depends on.

A strong post. Use a number from the explorer rather than a description of the shape. The rate at which the prior washes out is itself the interesting quantity, and a strong post treats it that way.

Week 5 · M5 Approximate Inference and Classification: no discussion

This week is the project midpoint (Part 1 is due), so there is no graded discussion. Run the [Week 5 logistic-regression tool](#) as lecture support; the comparison you would have written here (what Laplace and Metropolis each buy you and cost you on the same posterior) is exactly the analysis milestone 2.1 asks for, so nothing is lost, it just lives in the project.

Discussion 6 · Week 6 · M6 SVM and Clustering

When clusters do and do not match labels

Prompt. k-means on the six non-blood-pressure features recovers the hypertension strata only modestly. Report your own between-cluster relative risk. Then argue: is a modest enrichment a failure of the clustering, a failure of the features, or the correct answer to the question you actually asked? Second, the blood pressure columns were deliberately held out of the clustering. Say what would have happened to your relative risk if they had not been, and explain why that larger number would have been worse rather than better.

A strong post. Distinguish “the algorithm did not work” from “there is no strong geometric structure here to find.” The second question is about leakage, not about performance, and a strong post says so plainly.

Discussion 7 · Week 7 · M7 Dimensionality Reduction

The course arc: loss, likelihood, posterior, approximation, discovery

Prompt. From your PCA biplot, name what PC1 appears to encode on this cohort and which feature loadings told you that. Then trace the arc this course followed: loss, then likelihood, then posterior, then approximation, then discovery. Name the one step in that sequence where your understanding of the problem changed most, and say exactly what changed it. Close with something that **every** lens missed, and say whether more data or a different method would have caught it.

A strong post. The closing question is the real one. One specific thing all five steps missed, with a reason, beats a summary of what each step does.

Grading

Each week’s loop is **Pass** or **Not-Yet**, consistent with the rest of the course. There are no partial points. A loop passes when all four stages are complete and both your post and your responses meet the quality bar above. Six loops carry a discussion; in Week 5 the loop is the tool and its mastery check alone.

Because the stages are time-bound and depend on teammates, loops are not revisable after the week closes. The category is weighted to absorb an occasional miss rather than to punish one.

Peer-engagement loops and discussions are worth **10%** of the course grade. See [Assessment](#) for how the components combine.